Department of ICT
Faculty of Technology
University of Ruhuna

Data Structures and Algorithms – ICT2113 **Level 2- Semester 1**

Laboratory Assignment 8

Objectives:

Understand the implementation of Merge and Quick sorting techniques using C.

1. Write C program to sort the data set using **Merge sort** algorithm.

```
void mergesort(int arr[],int min ,int mid ,int max );
void partition(int arr[], int min, int max );

int main()
{
 int arr[30];
 int i,size;
 printf("\n\t------- Merge sorting -------\n\n");
 printf("Enter total no. of elements : ");
 scanf("%d",&size);

 for(i=0; i<size; i++)
 {
   printf("Enter %d element : ",i+1);
   scanf("%d",&arr[i]);
 }
 partition(arr,0,size-1);

 printf("\n\t------- After Merge sorted-------\n\n");
 for(i=0; i<size; i++)
 printf("%d ",arr[i]);

 return 0;
}
```

2. Write a program to implement the **quick sort** algorithm in C, display the sorted on the screen. Write a separate function for the quick sorting process.
   Use following function prototype

   void quicksort (int [15], int, int)